Identify airlines with significant delays to Las Vegas

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Introduction:

Air travel plays a pivotal role in modern life, connecting us to global destinations with remarkable convenience. However, the shadow of flight delays looms large, threatening to disrupt travel plans and cause frustration. For individuals contemplating flights to the dynamic city of Las Vegas, the ability to discern which airlines are more prone to significant delays becomes a critical factor in making informed travel choices.

Our research seeks to illuminate the prevalence of flight delays when journeying to Las Vegas. Central to our investigation is a rich dataset that encompasses crucial information, including carrier codes, flight dates, flight numbers, tail numbers, origin airports, scheduled and actual arrival times, scheduled and actual elapsed times, and arrival delays.

At the core of our research is a pivotal question: which airlines consistently experience the most substantial delays when flying to Las Vegas? By harnessing data-driven insights, we aim to empower travelers with the knowledge necessary to make well-informed decisions about their choice of flights to this iconic destination.

Our ultimate objective is to distill our findings into visually engaging presentations. These visualizations will serve as valuable resources, enabling travelers to optimize their experience when journeying to Las Vegas, aligning their choices with preferences and schedules.

Research Question:
Which airlines encounter the most substantial flight delays when flying to Las Vegas, and how can visualizing this data assist travelers in making informed flight choices?

Data Collection:
We collected data from “Kaggle” which encompasses records from the year 1987 to 2023. This dataset will help us uncover and comprehend valuable insights and trends that have extended over the years in the realm of Las Vegas-bound flights.

Data Preprocessing:

Data Selection:
The years 1987 to 2023 were thoughtfully selected as the data collection period to emphasize recent and pertinent data. This choice ensures alignment with our research objectives and underscores the data's relevance and quality.

Handling Missing Data:
After reviewing the dataset, we found no outliers, missing values, or instances of insufficient data. This robustness in the data underscores the quality and completeness of information, allowing us to conduct a thorough analysis and draw reliable insights

Data Format Conversion:
To optimize the management of date-related fields, such as 'date_(mm/dd/yyyy),' 'scheduled_arrival_time,' and 'actual_arrival_time,' we harnessed the power of Python's pandas library. The pd.to_datetime function facilitated a seamless conversion of these fields into the correct datetime format. Each field's format was meticulously specified, ensuring precision and clarity.

Data Analysis & Visualization:

The heart of our research lies in the analysis and visualization of this rich dataset. Through rigorous examination, patterns, and trends emerged, allowing us to draw insightful conclusions. Visualizations, such as histograms and box plots, illuminated the distribution of departure delays, airline-specific variations, and time-of-day trends. The data-driven insights we derived become actionable intelligence for travelers.

Business Value:

Our research brings actionable intelligence to the forefront. By unveiling the airlines that experience the most substantial delays when traveling to Las Vegas, we equip travelers with invaluable knowledge. With this information, individuals can make data-backed decisions when selecting flights, reducing travel-related uncertainties and enhancing their overall experience.

Not just the individuals, airlines can also use this information to improve their business operations and ensure more on-time flights, which will help retain customers for the long run.This not only ensures a more punctual service but also directly impacts customer loyalty and retention. Travelers will more likely choose airlines with consistent on-time performance, and by addressing these delays, airlines can bolster their competitive edge and cultivate a loyal customer base.

Beyond the immediate benefits for travelers and airlines, our research has far-reaching implications for the tourism and hospitality industry in Las Vegas. Timely and reliable flight arrivals are vital for the smooth functioning of the city's tourism sector, as they directly influence the schedules and plans of countless tourists. By providing valuable insights into which airlines boast the best on-time performance, our research contributes to a more efficiently functioning tourism ecosystem in Las Vegas.

Data Analytics :

1. Average delay by Airline, Time of Day, and Airport

In Figure 1.1 below, we have visualized the airlines along with their corresponding average delay times. Notably, the data reveals that Atlantic Southeast Airlines, formally known as ExpressJet, experiences the most substantial average delays (Atlantic Southeast Airlines had an average delay time of 14.6 minutes) , while Hawaiian Airlines stands out with the shortest average delay (3 minutes). This information can be pivotal for travelers seeking to minimize their wait times when selecting their preferred airline for flights to Las Vegas.

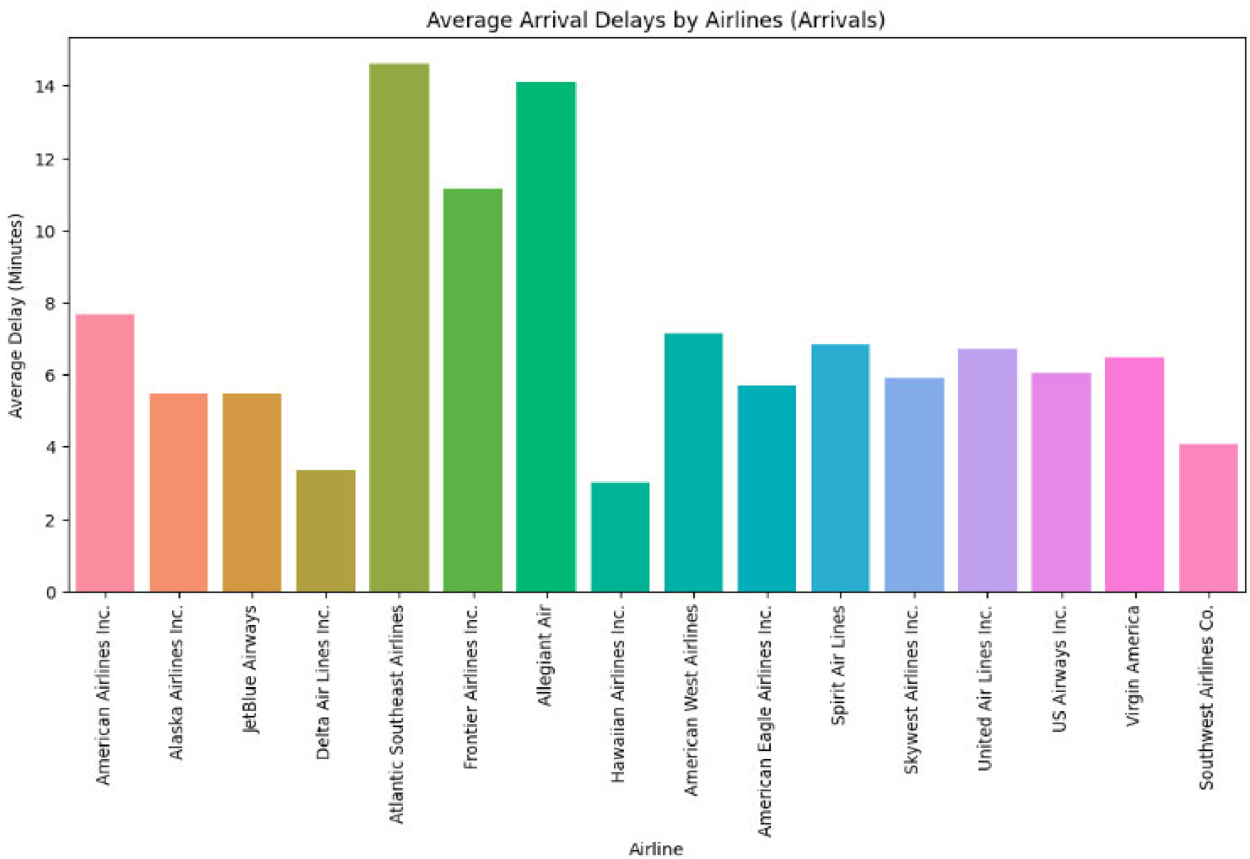

Figure 1.1: Average Delay by Airline

In our time-of-day analysis of arrival delays (Figure 1.2), we have uncovered a distinct pattern in the data. Morning flights consistently exhibit the least delay, making them the most reliable option for travelers heading to Las Vegas. The dataset analysis has unveiled some exciting insights about flight punctuality. Morning flights generally experience fewer delays compared to midday and evening flights. What's particularly noteworthy is that flights arriving in Las Vegas during the night are notably more dependable. These night flights often reach their destination ahead of their scheduled arrival time. This suggests that if you choose a night flight, you have a much better chance of arriving in Las Vegas more swiftly compared to flights during other times of the day.

One possible explanation for the punctuality of night flights is that they tend to have fewer passengers. Many travelers may find night travel less convenient, leading to reduced airport traffic during these hours. Additionally, night flights might be intentionally scheduled to depart on time or even ahead of schedule, contributing to their reliability.

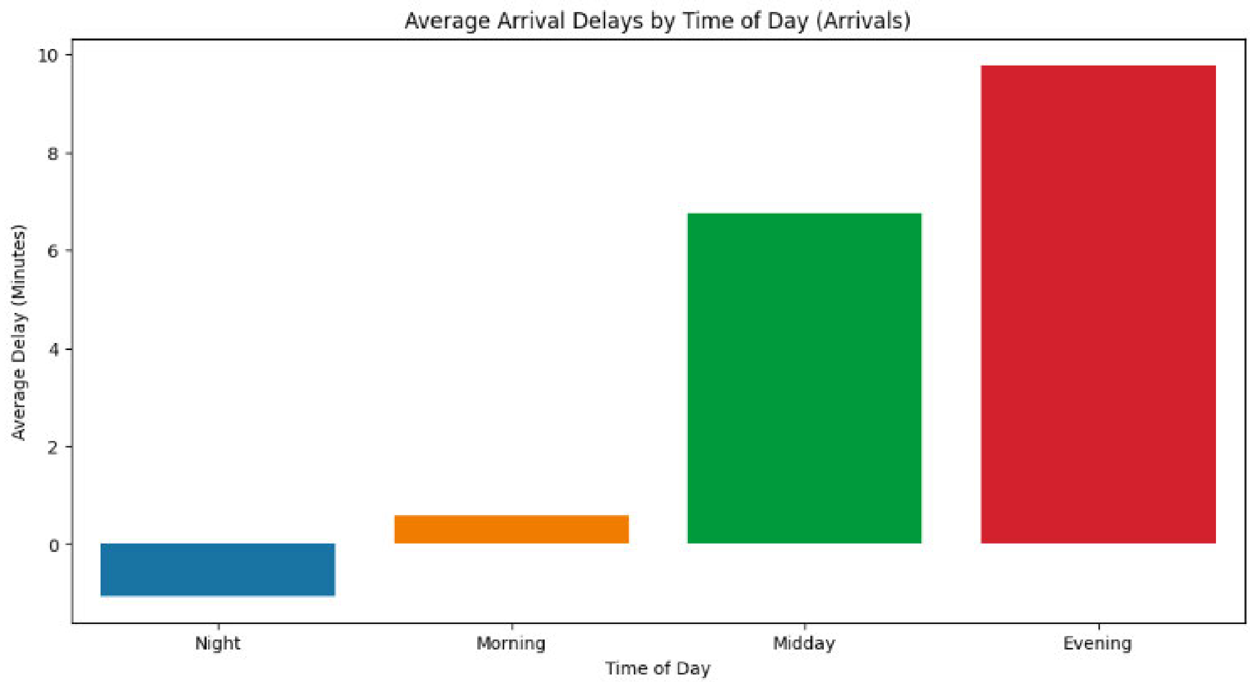

Figure 1.2: Average Delay by Time of Day

Throughout the years covered in the dataset, we've analyzed the top 10 delayed arrival flights to Las Vegas at National County International Airport. In contrast, based on Figure 1.3, Santa Maria Public Airport records the least delays. This data reveals that choosing National County International Airport for arriving in Las Vegas might not be the most optimal travel choice, as it has the highest delays among the airports considered in the analysis.

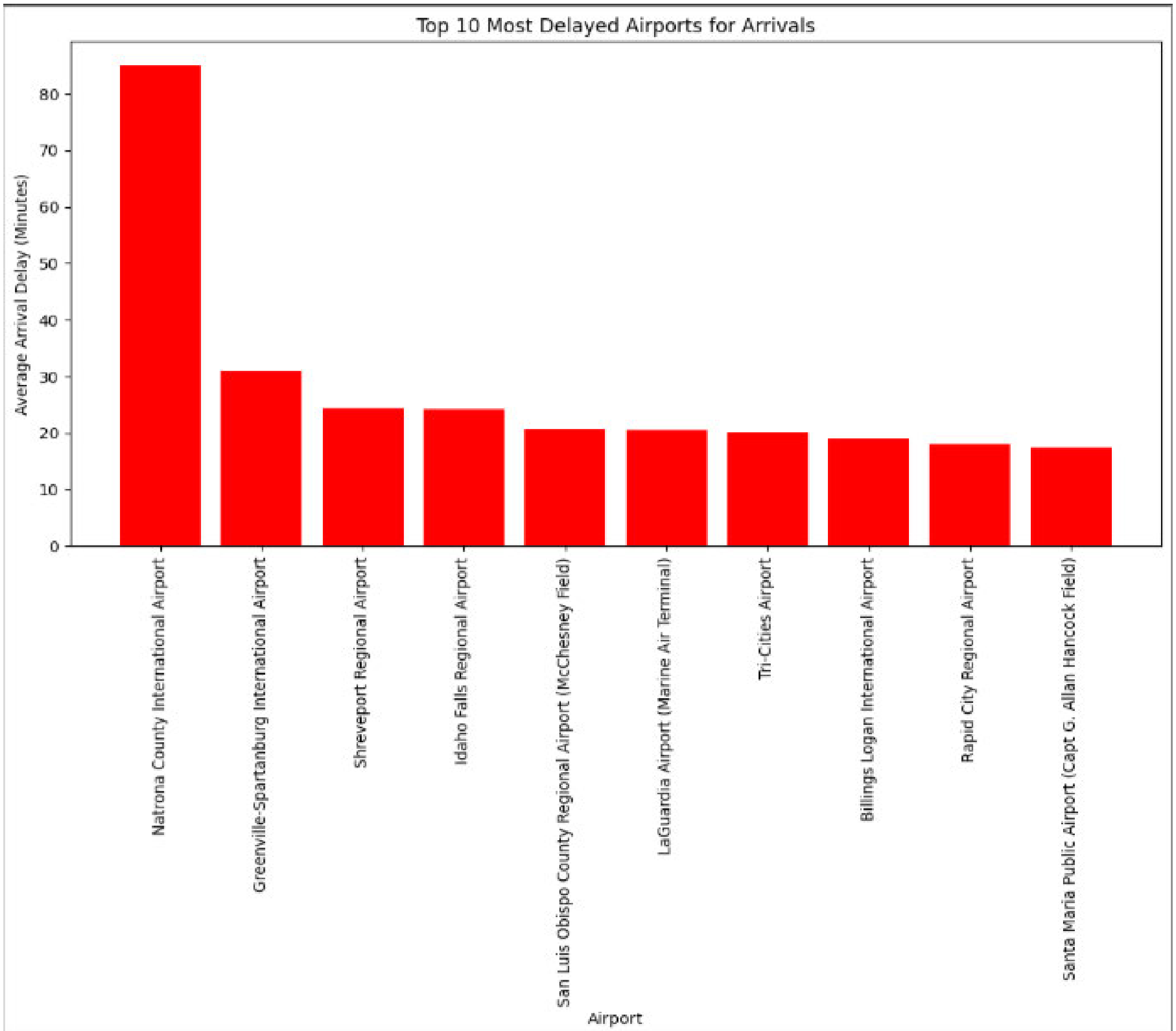

Figure 1.3: 10 Most Delayed Airport

2 . Delay for Arrival flights (in minutes)

Figure 2 below offers a comprehensive view of the distribution of various delay types for Arrival flights. Notably, the data highlights that the most prominent delay type is 'Late Aircraft Arrival,'(43.7%) constituting a significant portion of overall delays. Conversely, 'Security Delay' (0.3%) is the least frequent delay, contributing to a comparatively smaller portion of the total delays. The reason why ‘Late Aircraft Arrival’ is the most common type of delay could be because of the interconnected nature of airline schedules. If an aircraft arrives late from a previous flight, it can trigger a domino effect, causing subsequent departure also to be delayed. Understanding these delay patterns is critical for optimizing travel planning and mitigating potential disruptions in arrival flights.

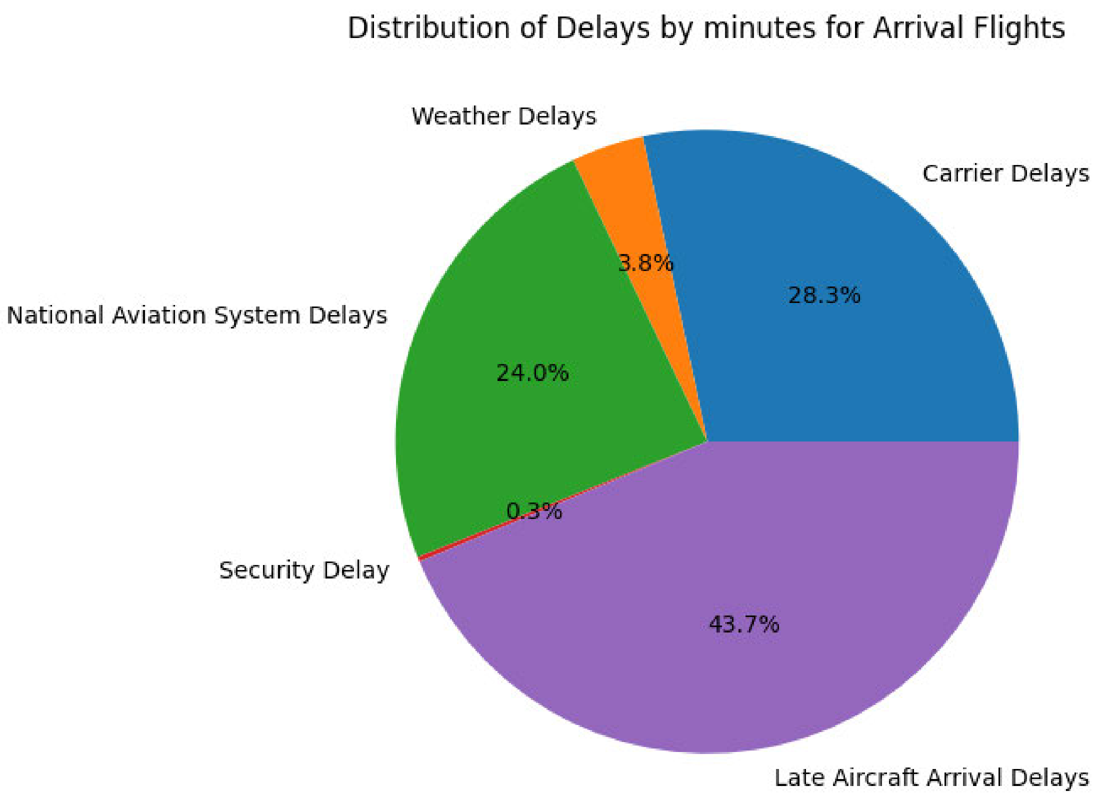

Figure 2: Delay for Arrival Flights

3. Average Arrival Delay (by years)

In this analysis, we delve into the trends associated with average arrival delays spanning multiple years. The process initiates with converting date entries into a uniform datetime format. Subsequently, the years are extracted from these date entries and incorporated into the dataset as a new 'year' column. The dataset is then grouped by year, and the mean arrival delay for each year is meticulously computed. To visually represent these trends, we've generated a line chart.

Figure 3.1 titled 'Average Arrival Delays Over the Years,' offers a valuable perspective on the evolution of average arrival delays over different years. Notably, the data underlines a noteworthy decrease in arrival delays in the year 2019, with this trend extending into the beginning of 2020. It's intriguing to observe that the most minor and most substantial delays occurred in recent years. However, it's crucial to acknowledge the exceptional circumstances of 2020. The lower delay in 2020 is likely attributable to the profound impact of the COVID-19 pandemic, which led to reduced air traffic and a decrease in overall flight operations during that period.

An important reason for the increase in travel delays after the COVID-19 pandemic can be attributed to the notable resurgence in travel. As the pandemic ended, people's eagerness to explore new destinations, reunite with family and friends, or resume business travel led to a rise in passenger demand. This increased demand for air travel put additional pressure on the aviation industry, contributing to delays. Moreover, during the COVID-19 pandemic, many airlines were forced to lay off personnel or implement cost-saving measures. When travel began to return to normal levels, there were potential challenges related to staffing and labor within the airlines. These challenges might include rehiring and retraining staff, addressing labor disputes, or simply adjusting to the sudden surge in operational demands, which could also contribute to flight delays.

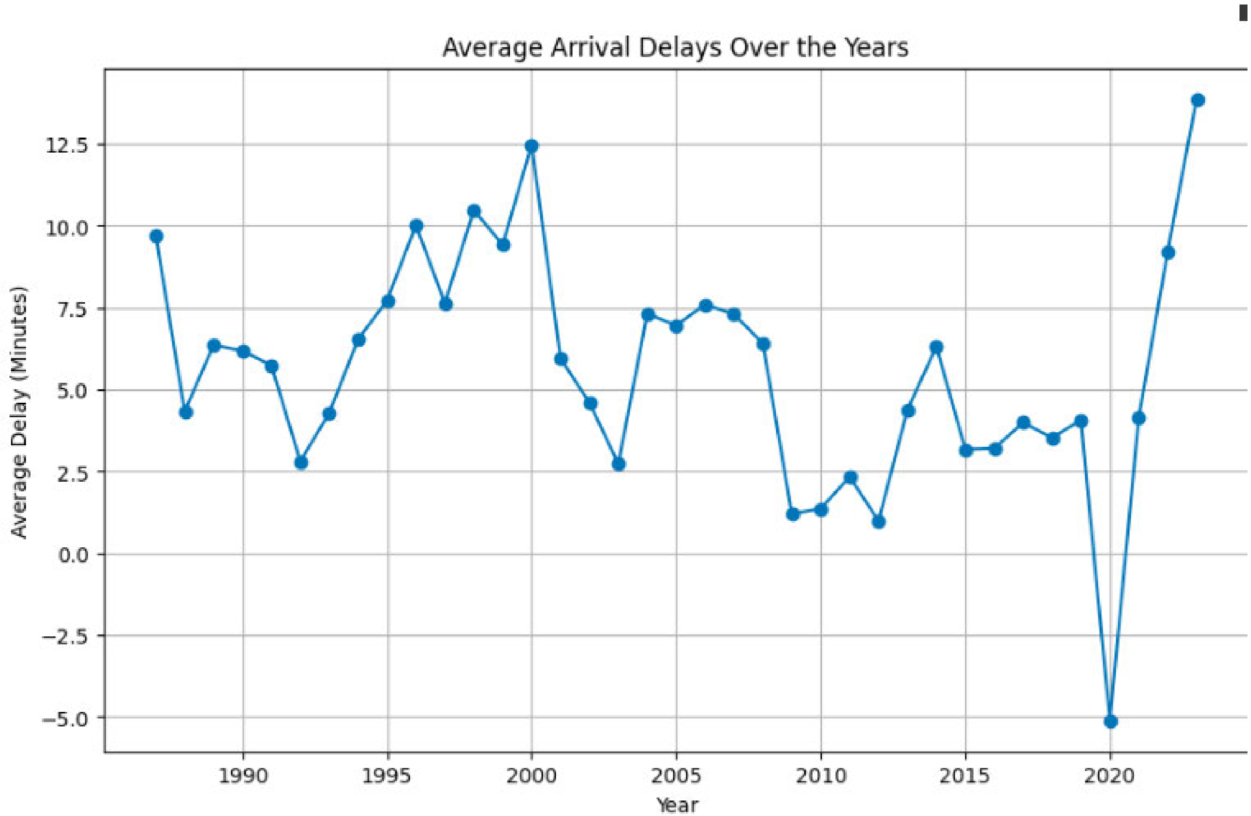

Figure 3.1: Delays by Year

In Figure 3.2, it's evident that the highest number of arrivals in Las Vegas occurs during March, while February records the fewest arrivals. The variations in arrival numbers can be attributed to several factors.

March is notable for having the highest number of arrivals in Las Vegas, and this trend can be linked to the occurrence of various spring holidays during this month. Many people visited Las Vegas in March to celebrate spring break, enjoy milder weather, and partake in various festivities, contributing to the increased arrivals.

Conversely, February registers the fewest arrivals in Las Vegas. One plausible explanation for this can be associated with February being traditionally a non-peak travel period. After the holiday season at the end of the previous year (December and January), many people tend to curb their travel plans in February, as they have recently returned from holiday trips and may be planning for spring or summer vacations later in the year.

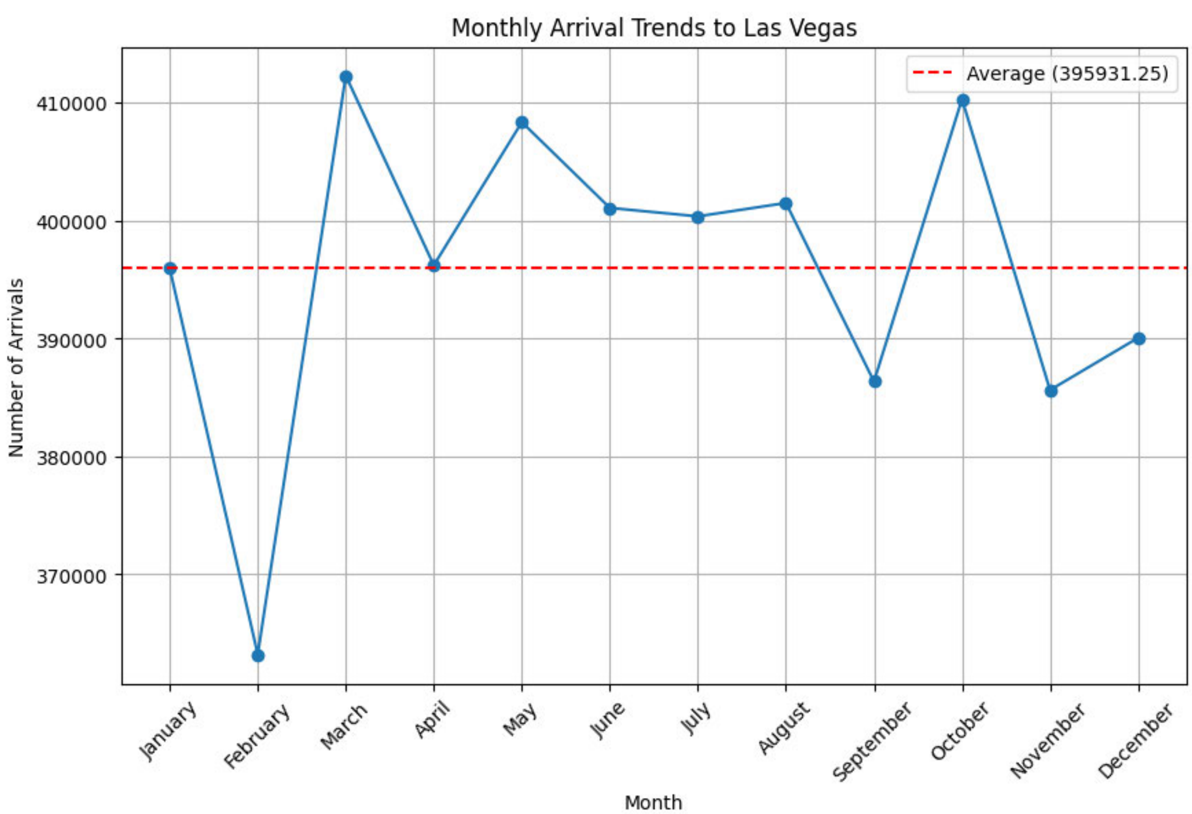

Figure 3.2: Monthly Arrival Trend

Conclusion:

In conclusion, for travelers heading to Las Vegas, selecting Hawaiian Airlines for Night flights can significantly reduce wait times. Conversely, Atlantic Southeast Airlines experiences the most substantial delays. Additionally, the analysis highlights the importance of considering airport choice, with National County International Airport having the highest delays, while Santa Maria Public Airport offers the most minor delays. Understanding delay patterns, such as 'Late Aircraft Arrival' being the most frequent delay, can aid in effective travel planning. The analysis also emphasizes the impact of the COVID-19 pandemic on recent delays and provides insights into monthly arrival trends, with March being the busiest month and February the quietest.

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